

MATH 20700 Honors Analysis I

Homework 5

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Problem 5

Proof. (a) Carries over. We know that the uniform limit of continuous functions is continuous.

- (b) Carries over. Suppose f has 11 discontinuities and fix ϵ . By uniform convergence, there exists an index N such that for all $n \geq N$, f_n is entirely within the ϵ band around f . Hence, all those f_n with $n \geq N$ must have discontinuities at the same 11 discontinuity points as f .

Alternatively, for each discontinuity i , let the size of the discontinuity be δ_i . Take $\delta = \min(\delta_1, \delta_2, \dots, \delta_{11})$. Let $\epsilon = \frac{\delta}{3}$. Then, since for all $n \geq N$, f_n is entirely within the ϵ band around f , all those f_n with $n \geq N$ must have discontinuities at the same 11 discontinuity points as f .

- (c) Does not carry over. Let $D = \{d_1, d_2, \dots, d_{11}\}$ be a fixed set of 11 points in the interval $[a, b]$ and define

$$f_n(x) = \begin{cases} 0, & \text{if } x \notin D_n \\ \frac{1}{n}, & \text{if } x \in D_n. \end{cases}$$

Then, clearly each f_n has 11 discontinuities. But $f_n \rightrightarrows f$ where f is the constant 0 function. For any $\epsilon > 0$, pick N such that $\frac{1}{N} < \epsilon$, then for all $n \geq N$, f_n is entirely within the ϵ band around the constant 0 function.

- (d) Does not carry over. Let's work on the interval $[a, b] = [0, 1]$.

First, let's design a limit function f that has infinitely many discontinuities.

Let's place a jump of size $1/2^k$ at each point $x = 1/k$. We can define this using a series:

$$f(x) = \sum_{k=1}^{\infty} \frac{1}{2^k} \cdot I(x \geq 1/k)$$

where I is the indicator function ($I(A) = 1$ if A is true, 0 if false). This function f has a jump discontinuity at every point $x \in \{1, 1/2, 1/3, \dots\}$, so it has infinitely many discontinuities.

We note that f_n is a finite sum of step functions. It is a step function with n jumps, located at the points $\{1, 1/2, 1/3, \dots, 1/n\}$. Since n is finite, f_n has finitely many discontinuities.

Let us check $f_n \rightrightarrows f$. We need to check the uniform norm, $\sup_{x \in [0,1]} |f(x) - f_n(x)|$.

$$|f(x) - f_n(x)| = \left| \sum_{k=1}^{\infty} \frac{1}{2^k} I(x \geq 1/k) - \sum_{k=1}^n \frac{1}{2^k} I(x \geq 1/k) \right| = \left| \sum_{k=n+1}^{\infty} \frac{1}{2^k} I(x \geq 1/k) \right|.$$

This “remainder” sum is always positive. Its maximum value (the supremum) occurs when $x = 1$, which makes all the $I(\dots)$ terms equal to 1.

$$\sup_{x \in [0,1]} |f(x) - f_n(x)| = \sum_{k=n+1}^{\infty} \frac{1}{2^k}.$$

This is a geometric series with sum $\frac{(1/2)^{n+1}}{1-1/2} = \frac{(1/2)^{n+1}}{1/2} = (1/2)^n = 1/2^n$. Since $\lim_{n \rightarrow \infty} (1/2^n) = 0$, the convergence is uniform.

Thus, we have a sequence f_n where each f_n has finitely many discontinuities (n of them), but the uniform limit f has infinitely many.

- (e) We denote D_n as the set of points where f_n is discontinuous, and D as the analogous set for f .

Our first assertion is that $D \subseteq \bigcup_{n \in \mathbb{N}} D_n$. To prove this, consider a point x that is not in the union $\bigcup D_n$. This choice of x means that every function f_n is continuous at x . A fundamental theorem states that uniform convergence preserves continuity. Therefore, the limit function f must also be continuous at x . This implies $x \notin D$. This confirms our assertion that $D \subseteq \bigcup_{n \in \mathbb{N}} D_n$.

By hypothesis, each D_n is a countable set. The union of a countable collection of countable sets remains countable. Since D is a subset of this countable union, D itself must be countable.

Next, we must demonstrate that f does not possess any oscillating discontinuities. We proceed by contradiction, assuming f does have an oscillating discontinuity at some point $t \in [a, b]$.

This implies that at least one of the one-sided limits fails to exist. Without loss of generality, let's assume the right-hand limit, $\lim_{x \rightarrow t^+} f(x)$, does not exist.

By the definition of a non-existent limit, there must be some $\epsilon > 0$ with the property that for any $\delta > 0$, we can find $x_1, x_2 \in (t, t + \delta) \cap [a, b]$ such that $|f(x_1) - f(x_2)| \geq \epsilon$.

The uniform convergence of f_n to f allows us to select an integer N such that for all x , $|f_N(x) - f(x)| < \epsilon/4$ (i.e., $\|f_N - f\|_{\infty} < \epsilon/4$).

A key premise is that f_N lacks oscillating discontinuities. Consequently, its right-hand limit at t , $\lim_{x \rightarrow t^+} f_N(x)$, must exist. The existence of this limit is equivalent to f_N satisfying the Cauchy criterion as $x \rightarrow t^+$. This means we can find a $\delta_N > 0$ such that for any pair of points $x, y \in (t, t + \delta_N) \cap [a, b]$, the inequality

$$|f_N(x) - f_N(y)| < \frac{\epsilon}{2}$$

holds.

Let us now apply our contradictory assumption on f within this specific interval $(t, t + \delta_N)$. Our assumption guarantees the existence of two points, let's call them x_0 and y_0 , within $(t, t + \delta_N) \cap [a, b]$ for which $|f(x_0) - f(y_0)| \geq \epsilon$.

We now examine the behavior of f_N at these two specific points, x_0 and y_0 , using the reverse triangle inequality:

$$\begin{aligned} |f_N(x_0) - f_N(y_0)| &\geq |f(x_0) - f(y_0)| - |f_N(x_0) - f(x_0)| - |f_N(y_0) - f(y_0)| \\ &\geq \epsilon - \frac{\epsilon}{4} - \frac{\epsilon}{4} = \frac{\epsilon}{2}. \end{aligned}$$

The result $|f_N(x_0) - f_N(y_0)| \geq \epsilon/2$ is in direct conflict with the Cauchy property of f_N , which established that $|f_N(x) - f_N(y)| < \epsilon/2$ for all x, y in this interval.

This contradiction invalidates our initial assumption. We must conclude that f has no oscillating discontinuities.

- (f) Assume $f_n \rightarrow f$ uniformly, and no f_n has jump discontinuities. Suppose f has a jump at c , with $\Delta := |f(c+) - f(c-)| > 0$. Let $\epsilon = \Delta/5$. Choose $\delta > 0$ such that for $0 < h < \delta$,

$$|f(c+h) - f(c+)| < \epsilon, \quad |f(c-h) - f(c-)| < \epsilon.$$

By uniform convergence, choose N so $\|f_N - f\|_\infty < \epsilon$. Then for $0 < h < \delta$, the triangle inequality gives

$$|f_N(c+h) - f(c+)| < 2\epsilon, \quad |f_N(c-h) - f(c-)| < 2\epsilon.$$

This implies the f_N values near c (from the right and left) are separated by at least $\Delta - 4\epsilon = \epsilon > 0$. This gap means f_N must have a jump, a contradiction. Thus, f has no jump discontinuities.

- (g) The property of having “no oscillating discontinuities” means that all one-sided limits, $f_n(x+)$ and $f_n(x-)$, exist for all x . Since $f_n \rightarrow f$ uniformly, it follows that the one-sided limits $f(x+)$ and $f(x-)$ also exist for all x . Therefore, f has no oscillating discontinuities.

QED

Problem 10

Proof. Define

$$f_n(x) = \begin{cases} 0 & \text{if } x \leq n \\ n(x - n) & \text{if } n < x < n + 1/n \\ 1 & \text{if } x \geq n + 1/n. \end{cases}$$

Uniformly Continuous: Each f_n is continuous and has a bounded derivative ($|f'_n(x)| \leq n$). This makes each f_n Lipschitz continuous, which implies it is uniformly continuous.

Pointwise Equicontinuous: To show equicontinuity at a fixed $x \in \mathbb{R}$, choose an integer $N > |x|$. For all $n \geq N$, f_n is identically zero in a neighborhood of x (e.g., $(x-1, x+1)$). We only need to consider the finite, uniformly continuous family $\{f_1, \dots, f_{N-1}\}$, for which a common $\delta > 0$ can always be found.

Not Uniformly Equicontinuous: The family fails uniform equicontinuity because the slopes n are unbounded. To prove this, let $\epsilon = 1/2$. For any $\delta > 0$, choose n large enough such that $1/n < \delta$. Let $x = n$ and $y = n + 1/n$. Then $|x - y| = 1/n < \delta$, but $|f_n(y) - f_n(x)| = |1 - 0| = 1 \geq \epsilon$.

QED

Problem 15

Proof. (a) First we want to show that if f is uniformly continuous then it has a modulus of continuity.

Fix $\epsilon > 0$, then by uniform continuity, $\exists \delta > 0$ such that if $|s - t| < \delta$ then $|f(s) - f(t)| < \epsilon$.

We know that $\bigcup_{x \in [a, b]} B_\delta(x)$ is an open cover for the interval $[a, b]$ so by compactness of the interval $[a, b]$, there exists a finite subcover. In other words, finitely many (say n -many) open balls of radius δ cover the interval, i.e., $[a, b] \subset \bigcup_{1 \leq i \leq n} B_\delta(m_i)$.

Then, any two points $s, t \in [a, b]$ are at most $n\delta$ apart, so for any two points $s, t \in [a, b]$, we have $|f(s) - f(t)| < n\epsilon$.

On each of the δ -balls, consider the variation of the function f , and take the supremum. In other words, let $y = \sup_{1 \leq i \leq n} (\sup_{j, k \in B_\delta(m_i)} |f(j) - f(k)|)$.

So we want to define $\mu : (0, \infty) \rightarrow (0, \infty)$ such that $\mu(\delta) \leq y$ so for simplicity let us say $\mu(\delta) = y$.

Let's define $\omega(x)$ as $\omega(x) = \sup_{t \in [a, b]} (\sup_{j, k \in [t, t+x] \cap [a, b]} |f(j) - f(k)|)$. We note that this is equivalent to $\omega(x) = \sup\{|f(s) - f(t)| : s, t \in [a, b], |s - t| \leq x\}$.

So defining $\mu(x) = x^2 + \omega(x)$ works, because it is easy to see that $\mu(s) \rightarrow 0$ as $s \rightarrow 0$ and μ is strictly increasing.

Next we want to show that if f has a modulus of continuity then it is uniformly continuous.

Fix $\epsilon > 0$. Since $\mu(s) \rightarrow 0$ as $s \rightarrow 0$, we have that $\exists y$ such that $\mu(y) < \epsilon$. Then choosing $\delta = y$ completes the proof for uniform continuity.

- (b) If the members have a common modulus of continuity μ , then fix $\epsilon > 0$. Since $\mu(s) \rightarrow 0$ as $s \rightarrow 0$, we have that $\exists y$ such that $\mu(y) < \epsilon$. Then choosing $\delta = y$ completes the proof for uniform continuity. Note that this δ works for all the members.

If the family of functions is equicontinuous, then there is a single δ that works for uniform continuity of all the members. In part (a) above, we defined $\omega_f(x) = \sup\{|f(s) - f(t)| : s, t \in [a, b], |s - t| \leq x\}$.

Let $\omega_{\mathcal{F}}(s) = \sup_{f \in \mathcal{F}} \{\omega_f(s)\}$ and let $\mu(s) = s + \omega_{\mathcal{F}}(s)$. Here, s is strictly increasing and $\omega_{\mathcal{F}}(s)$ is non-decreasing, hence μ is strictly increasing.

For any $f \in \mathcal{F}$ and any $s, t \in [a, b]$, we have $|f(s) - f(t)| \leq \omega_f(|s - t|) \leq \sup_{g \in \mathcal{F}} \{\omega_g(|s - t|)\} = \omega_{\mathcal{F}}(|s - t|)$. Since $|s - t| \geq 0$, this implies that $|f(s) - f(t)| \leq \omega_{\mathcal{F}}(|s - t|) + |s - t| = \mu(|s - t|)$.

We want to show that $\omega_{\mathcal{F}}(s) \rightarrow 0$ as $s \rightarrow 0$. By equicontinuity, given any $\epsilon > 0$, there exists a $\delta > 0$ such that for all $f \in \mathcal{F}$, if $|j - k| < \delta$, then $|f(j) - f(k)| < \epsilon$. This implies that for any $s' < \delta$, $\omega_f(s') = \sup\{|f(j) - f(k)| : |j - k| \leq s'\} \leq \epsilon$ for all $f \in \mathcal{F}$. Therefore, $\omega_{\mathcal{F}}(s') = \sup_{f \in \mathcal{F}} \{\omega_f(s')\} \leq \epsilon$ for all $s' < \delta$. Thus, $\omega_{\mathcal{F}}(s) \rightarrow 0$ as $s \rightarrow 0$ so clearly $\mu(s) \rightarrow 0$ as $s \rightarrow 0$.

QED

Problem 17

Proof. (a) We know from 15 (a) that a family of functions is equicontinuous if and only if its members have a common modulus of continuity.

$\forall f \in \mathcal{F}$, we know that $H_f \leq H$. Hence, $\mu_f(s) = H_f s^\alpha \leq H s^\alpha := \mu^*(s)$.

Thus, $\forall g \in \mathcal{F}$, we have $|g(s) - g(t)| \leq \mu_g(|s - t|) \leq \mu^*(|s - t|)$. Thus, μ^* is a common modulus of continuity.

(b) Yes, same thing is true. It is easy to see this by cases:

1. $s, t \in (a, b)$
2. $s = a, t \in (a, b)$
3. $s = b, t \in (a, b)$

In neither case is the core argument affected.

(c) Yes, same thing is true. Here, it helps to make explicit the single δ that works for uniform continuity of all $f \in \mathcal{F}$.

From part (a), $\forall g \in \mathcal{F}$, we have $|g(s) - g(t)| \leq \mu_g(|s - t|) \leq \mu^*(|s - t|) < \epsilon$, where the last inequality is the one we desire for uniform continuity.

Thus, we want $H|s - t|^\alpha < \epsilon$, so $|s - t| < (\frac{\epsilon}{H})^{1/\alpha}$.

Thus, $\delta = (\frac{\epsilon}{H})^{1/\alpha}$ works for uniform continuity of all $f \in \mathcal{F}$. Note that this δ does not depend at all on the specific location of the points s and t : as long as they are δ close, their function values will be $\mu^*(|s - t|)$ close.

(d) Yes, same thing is true because $\mathbb{Q} \subset \mathbb{R}$ so the core argument involving a single uniform δ is not affected.

(e) In this case, picking $\delta = 0.5$ guarantees uniform equicontinuity, because $|s - t| < \delta = 0.5$ can only hold for $s, t \in \mathbb{N}$ if $s = t$, hence $f(s) = f(t)$.

QED

Problem 18

Proof. (a) We know that $\forall n, |f_n(p)| \leq P$ and we want to show that if $x \in [a, b], x \neq p$, then $|f_n(x)| \leq X$, i.e., it is bounded.

$$|f_n(x)| = |f_n(x) - f_n(p) + f_n(p)| \leq |f_n(x) - f_n(p)| + |f_n(p)| \leq |f_n(x) - f_n(p)| + P.$$

So it suffices to show a bound for $|f_n(x) - f_n(p)|$. Let us fix $\epsilon > 0$ and call the δ for uniform equicontinuity as δ_{equi} .

If $|x - p| < \delta_{\text{equi}}$ then $|f_n(x) - f_n(p)| < \epsilon$ and we are done.

Let $J = \lceil |b - a|/\delta_{\text{equi}} \rceil$. This J is a fixed integer that only depends on the interval and ϵ , not on x or n . Since $|x - p| \leq |b - a|$ for any x , we can always bridge the gap from p to x in at

most J steps, each smaller than δ_{equi} . By repeated use of the triangle inequality, this gives $|f_n(x) - f_n(p)| < J \cdot \epsilon$.

This proves that $|f_n(x)|$ is bounded. x was arbitrary, hence (f_n) is uniformly bounded. To make it explicit, the uniform bound is $|f_n(x)| \leq J \cdot \epsilon + P$.

(b) If (f_n) is an equicontinuous sequence in C^0 and there exists $p \in [a, b]$ such that $(f_n(p))$ is a bounded sequence of real numbers, then there exists a uniformly convergent subsequence of (f_n) .

- (c) 1. Yes, (a, b) works. This is because the bound from part (a) works exactly the same.
 2. No, $\mathbb{Q}$ does not work. Same logic as the case of $\mathbb{R}$ (see below).
 3. No, $\mathbb{R}$ does not work.

In our proof for $[a, b]$ in part (a), we found a single, finite number of steps J (which depended on $|b - a|$) to get from p to any x . If our domain is now $\mathbb{R}$, we can no longer find one finite J that works for all $x \in \mathbb{R}$, since there could be points arbitrarily far from p .

4. For any $\epsilon > 0$, choosing $\delta = 0.5$ makes the equicontinuity condition $|x - y| < \delta \implies |f_n(x) - f_n(y)| < \epsilon$ vacuously true, because no two distinct x, y in $\mathbb{N}$ satisfy $|x - y| < 0.5$. This means every sequence of functions on $\mathbb{N}$ is equicontinuous.

So, does boundedness at a single point p imply uniform boundedness?

No. We construct a simple sequence $f_n : \mathbb{N} \rightarrow \mathbb{R}$ that is bounded at $p = 1$ but is not uniformly bounded (we define $f_n(j) = j^n$). Then, $f_n(1) = 1$ for all n , but the sequence is not uniformly bounded.

(d) Generalization of part (a): The theorem “an equicontinuous sequence that is bounded at one point is uniformly bounded” holds if and only if the domain K is a bounded metric space.

Generalization of part (b): The theorem “an equicontinuous sequence that is bounded at one point has a uniformly convergent subsequence” holds if and only if the domain K is a compact metric space.

QED

Problem 21

Proof. (a) Fix $\epsilon > 0$ and $x \in [a, b]$. Then, we want to show that $\exists \delta > 0$ such that if $|x - y| < \delta$ then $|\sup_{f \in \mathcal{E}} f(y) - \sup_{f \in \mathcal{E}} f(x)| < \epsilon$.

Let us define $g(x) = \sup_{f \in \mathcal{E}} f(x)$.

By uniform equicontinuity of $\mathcal{E}$, $\exists \delta > 0$ such that if $|x - y| < \delta$ then $\forall f \in \mathcal{E}, |f(x) - f(y)| < \epsilon$.

Thus, $\forall f \in \mathcal{E}, f(y) < f(x) + \epsilon \leq g(x) + \epsilon$. Thus, $g(y) = \sup_{f \in \mathcal{E}} f(y) \leq g(x) + \epsilon$.

Similarly, $\forall f \in \mathcal{E}, f(x) < f(y) + \epsilon \leq g(y) + \epsilon$, so taking the supremum over f gives $g(x) \leq g(y) + \epsilon$.

Thus, we conclude that if $|x - y| < \delta$ then $|g(x) - g(y)| \leq \epsilon$. This shows that g is continuous.

In fact, we have shown that g is uniformly continuous. Simple continuity of g would hold even if $\mathcal{E}$ was pointwise equicontinuous instead of uniformly equicontinuous.

- (b) For a counterexample, consider the sequence (f_n) with $f_n : [-1, 1] \rightarrow \mathbb{R}$ defined as $f_n(x) = \begin{cases} 0, & \text{if } x \leq 0 \\ \min\{1, nx\}, & \text{if } x > 0. \end{cases}$

Clearly, each f_n is continuous and bounded.

Let's check that it fails equicontinuity. At $x_0 = 0$ with $\epsilon = 1/2$, for any $\delta > 0$, we can choose n large enough such that $x = 1/n < \delta$. Then $|x - 0| < \delta$, but $|f_n(x) - f_n(0)| = |f_n(1/n) - 0| = |\min\{1, n(1/n)\}| = \min\{1, 1\} = 1$. Since $1 \not< \epsilon$, the sequence is not equicontinuous at $x = 0$.

Let's find $g(x) = \sup_{f \in (f_n)} f(x)$. If $x \leq 0$, $g(x) = \sup\{0\} = 0$. If $x > 0$, $g(x) = \sup_n \{\min\{1, nx\}\}$. For any fixed $x > 0$, we can find an n large enough (specifically, $n \geq 1/x$) so that $nx \geq 1$. For that n , $f_n(x) = 1$. The supremum must therefore be 1. So,

$$g(x) = \begin{cases} 0, & \text{if } x \leq 0 \\ 1, & \text{if } x > 0. \end{cases}$$

This function $g(x)$ is not continuous at $x = 0$.

- (c) For a counterexample, consider the sequence (f_n) with $f_n : [0, 1] \rightarrow \mathbb{R}$ defined as $f_n(x) = x^n$.

We know that (f_n) is not equicontinuous at $x = 1$, because if we pick $\epsilon = 1/2$, then for any $\delta > 0$, we can choose a point x (like $x = 1 - \delta/2$) close to 1 but with $x < 1$. Since $x^n \rightarrow 0$ for this x , we can find an n large enough so that $x^n < 1/2$, which means $|f_n(x) - f_n(1)| = |x^n - 1| = 1 - x^n > 1/2 = \epsilon$, violating the condition.

We want to show that the sup is continuous. Let $g(x) = \sup_n f_n(x)$. We observe that $g(0) = 0$ because $\forall n, f_n(0) = 0^n = 0$. Similarly, $g(1) = 1$ because $\forall n, f_n(1) = 1^n = 1$.

For $m \in (0, 1)$, we observe that $\forall n, f_{n+1}(m) = m^{n+1} < m^n = f_n(m)$. Hence, $\forall m \in (0, 1), g(m) = f_1(m) = m^1 = m$.

Thus, we have $g(0) = 0, g(x) = x$ for $x \in (0, 1)$ and $g(1) = 1$. We can rewrite this as $g(x) = x$ for all $x \in [0, 1]$ which is clearly continuous.

- (d) Same counterexample as part (c) above. Let $\mathcal{E}$ be the sequence of functions (f_n) defined in part (c), then any subset must be of the form $\{x^n : n \in \mathcal{N}\}$ for some collection of natural numbers $\mathcal{N} \subseteq \mathbb{N}$. Then clearly, by the same argument as part (c) above, the supremum $g(x) = x^k$ where $k = \inf \mathcal{N}$.

QED

Problem 24

Proof. (a) Let $g_n(x) = f_n(x) - f(x)$. Our goal is to prove that $f_n \rightarrow f$ uniformly, which is equivalent to proving $g_n \rightarrow 0$ uniformly. This new sequence g_n has the following properties:

1. Continuous: Each g_n is continuous (as it's the difference of two continuous functions).
2. Monotonic: $g_n(x) \geq g_{n+1}(x)$ (since f_n is decreasing).

3. Non-negative: Since the sequence $f_n(x)$ decreases to $f(x)$, we have $f_n(x) \geq f(x)$, which means $g_n(x) \geq 0$.
4. Pointwise Convergence: $g_n(x) \rightarrow 0$ for every $x \in [a, b]$.

Fix an $\epsilon > 0$. For each $x \in [a, b]$, since $g_n(x) \rightarrow 0$, there exists an integer N_x (which may depend on x) such that $g_{N_x}(x) < \epsilon$.

Since g_{N_x} is continuous, there exists an open neighborhood U_x around x such that for all $y \in U_x \cap [a, b]$, we also have $g_{N_x}(y) < \epsilon$.

The collection of these open sets, $\{U_x\}_{x \in [a, b]}$, forms an open cover of the compact set $[a, b]$.

By the Heine-Borel theorem, this open cover has a finite subcover. That is, there exists a finite set of points $\{x_1, \dots, x_k\}$ such that $[a, b] \subseteq U_{x_1} \cup \dots \cup U_{x_k}$.

Let $N = \max\{N_{x_1}, \dots, N_{x_k}\}$. This is a single integer that does not depend on x . Now, take any $y \in [a, b]$. Since the U_{x_i} form a cover, y must belong to some U_{x_j} in this finite collection.

By the construction of U_{x_j} , we know that $g_{N_{x_j}}(y) < \epsilon$. By our choice of N , we know $N \geq N_{x_j}$. By the monotonic decreasing property of g_n , this means $g_N(y) \leq g_{N_{x_j}}(y)$. Combining these, we get $g_N(y) \leq g_{N_{x_j}}(y) < \epsilon$.

Since y was arbitrary, this holds for all $y \in [a, b]$. Therefore, for any $n \geq N$, we have $0 \leq g_n(x) \leq g_N(x) < \epsilon$ for all $x \in [a, b]$. This is the definition of uniform convergence $g_n \rightarrow 0$.

- (b) Switching g to be defined as $g(x) = f(x) - f(x_n)$, the exact same argument as part (a) holds, hence the convergence is uniform even in the increasing case.
- (c) $\mathbb{R}$ is not compact, hence the argument of part (a) fails. ($f_n : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f_n(x) = \arctan(x - n)$ is a specific counterexample).
- (d) Replacing $[a, b]$ with a compact metric space keeps the argument of part (a) intact. $\mathbb{R}^m$ is not compact, hence the argument of part (a) fails. ($f_n : \mathbb{R}^m \rightarrow \mathbb{R}$ defined as $f_n(x) = \arctan(x_1 - n)$ is a specific counterexample).

QED

Problem 35

Proof. First, if $x_0 = 0$, the unique solution is the constant function $x(t) = 0$, which is defined for all time.

For $x_0 \neq 0$, we can solve the ODE $dx/dt = x^2$ by separating variables: $\int x^{-2} dx = \int dt$. Integrating both sides yields $-1/x = t + C$. Using the initial condition $x(0) = x_0$, we find the constant $C = -1/x_0$. Substituting C back and solving for $x(t)$ gives the solution $x(t) = \frac{1}{1/x_0 - t}$.

This solution is not defined for all time. It exhibits a "finite-time blow-up" because the denominator $1/x_0 - t$ becomes zero when $t = 1/x_0$. As t approaches this finite time, the solution $x(t)$ escapes to infinity.

QED

Problem 36

Proof. (a) Pick any arbitrary finite time t , then we want to show that $x(t) < \infty$.

By Mean Value Theorem, $|x(t) - x(0)| = |x'(y)| \cdot |t| = |f(x(y))| \cdot |t|$ for some y between 0 and t .

Since $|f(x(y))| \leq M$, we get $|x(t) - x(0)| \leq M|t|$.

By the triangle inequality, $|x(t)| - |x(0)| \leq |x(t) - x(0)| \leq M|t|$, which gives $|x(t)| \leq |x(0)| + M|t| < \infty$ for finite time.

(b) By Fundamental Theorem of Calculus,

$$\begin{aligned} |x(t)| &= |x(0) + \int_0^t f(x(s)) ds| \\ &\leq |x(0)| + \left| \int_0^t f(s) ds \right| \\ &\leq |x(0)| + \int_0^t |f(x(s))| ds \\ &\leq |x(0)| + \int_0^t C|x(s)| ds + \int_0^t K ds \\ &= |x(0)| + C \int_0^t |x(s)| ds + \int_0^t K ds \\ &= g(t). \end{aligned}$$

Above, we define $g(t) = |x(0)| + C \int_0^t |x(s)| ds + \int_0^t K ds$, then we observe that $g'(t) = C|x(t)| + K$.

We note that $|x(t)| \leq g(t)$ from the above chain of inequalities, hence $g'(t) \leq Cg(t) + K$.

Let us investigate the related differential equation $h'(t) = Ch(t) + K$. We note that $h' = C(h + K/C)$, hence $\int \frac{1}{K/C+h} dh = \int C dt + j$ where j is the integration constant. Thus, $\ln(h + \frac{K}{C}) = Ct + j$. Thus, $h + \frac{K}{C} = e^{Ct+j}$ hence $h(t) = -\frac{K}{C} + e^{Ct+j} = -\frac{K}{C} + (h(0) + \frac{K}{C})e^{Ct}$.

Thus, $g(t) \leq h(t) < \infty$ for finite time.

(c) Replace absolute value $||$ with the Euclidean metric $|| \cdot ||$ everywhere, then the proofs of parts (a) and (b) carry over intact.

(d) First, we prove that uniform continuity implies the growth condition $||f(x)|| \leq C ||x|| + K$.

By uniform continuity, for $\epsilon = 1$, there exists a $\delta > 0$ such that for any $y, z \in \mathbb{R}^m$, $||y - z|| < \delta \implies ||f(y) - f(z)|| < 1$.

For any $x \in \mathbb{R}^m$ (assuming $x \neq 0$), let $n = \lfloor \frac{||x||}{\delta} \rfloor + 1$. This integer n satisfies $n > \frac{||x||}{\delta}$ (which implies $\frac{||x||}{n} < \delta$) and $n \leq \frac{||x||}{\delta} + 1$.

Define a sequence of points $v_i = \frac{i}{n}x$ for $i = 0, \dots, n$. This creates a chain from $v_0 = 0$ to $v_n = x$ where the distance between adjacent points is $||v_i - v_{i-1}|| = \frac{||x||}{n} < \delta$.

Now, we bound $\|f(x)\|$ using the triangle inequality on the telescoping sum $f(x) - f(0) = \sum_{i=1}^n (f(v_i) - f(v_{i-1}))$:

$$\|f(x)\| \leq \|f(x) - f(0)\| + \|f(0)\| \leq \sum_{i=1}^n \|f(v_i) - f(v_{i-1})\| + \|f(0)\|$$

Since each step $\|v_i - v_{i-1}\| < \delta$, each term in the sum is < 1 . Thus:

$$\|f(x)\| < n + \|f(0)\|$$

Substituting our upper bound for n :

$$\|f(x)\| < \left(\frac{\|x\|}{\delta} + 1 \right) + \|f(0)\| = \left(\frac{1}{\delta} \right) \|x\| + (1 + \|f(0)\|)$$

This is precisely the condition $\|f(x)\| \leq C \|x\| + K$ with constants $C = 1/\delta$ and $K = 1 + \|f(0)\|$.

Inference: We have just shown that if f is uniformly continuous, then the condition stated in part (b) is true. By the result of part (c), any solution to $x' = f(x)$ in $\mathbb{R}^m$ satisfying this condition does not escape to infinity in finite time. Therefore, solutions of uniformly continuous ODEs defined on $\mathbb{R}^m$ do not escape to infinity in finite time.

QED

Problem 43

Proof. It is a reference to a poem!

QED